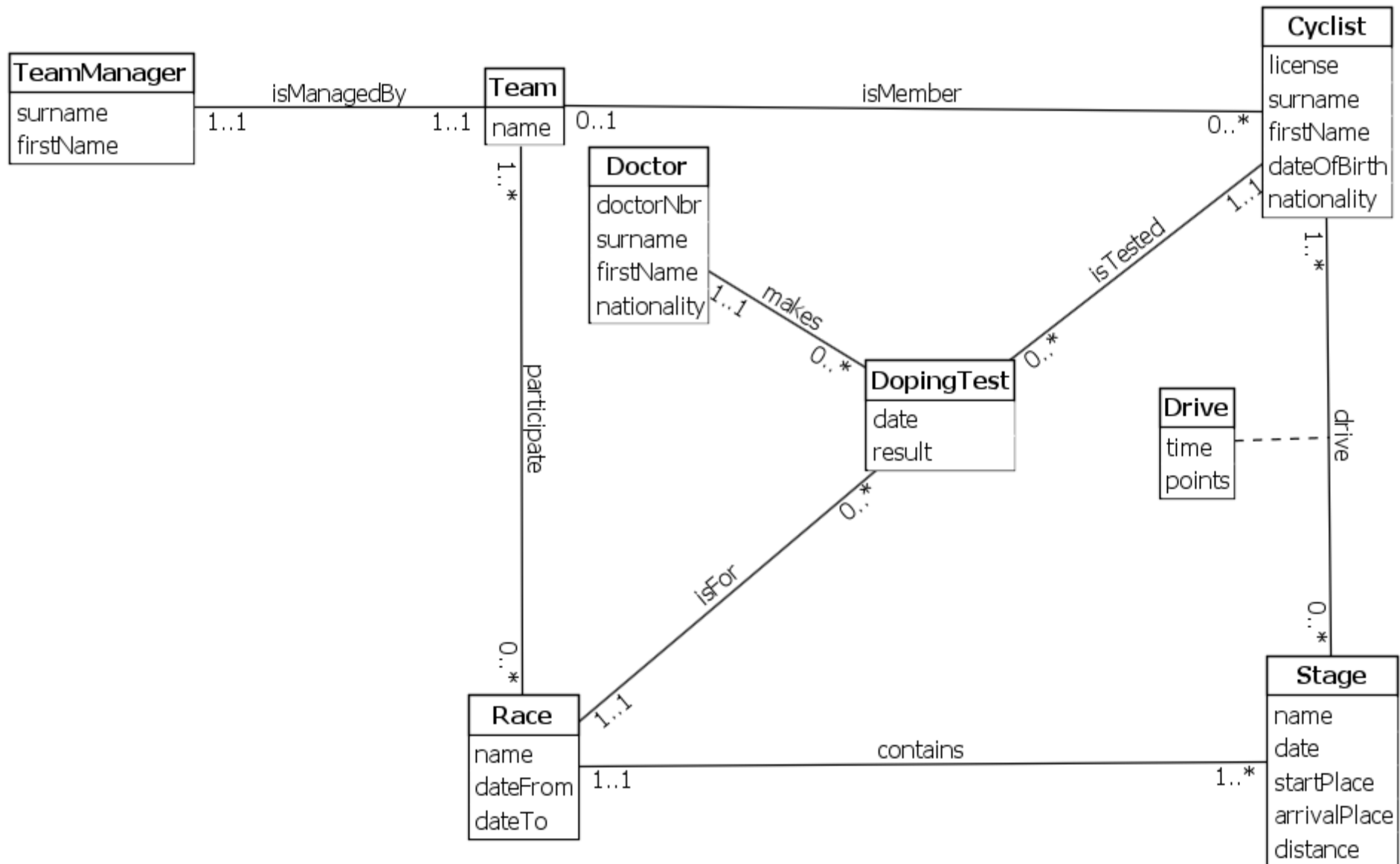
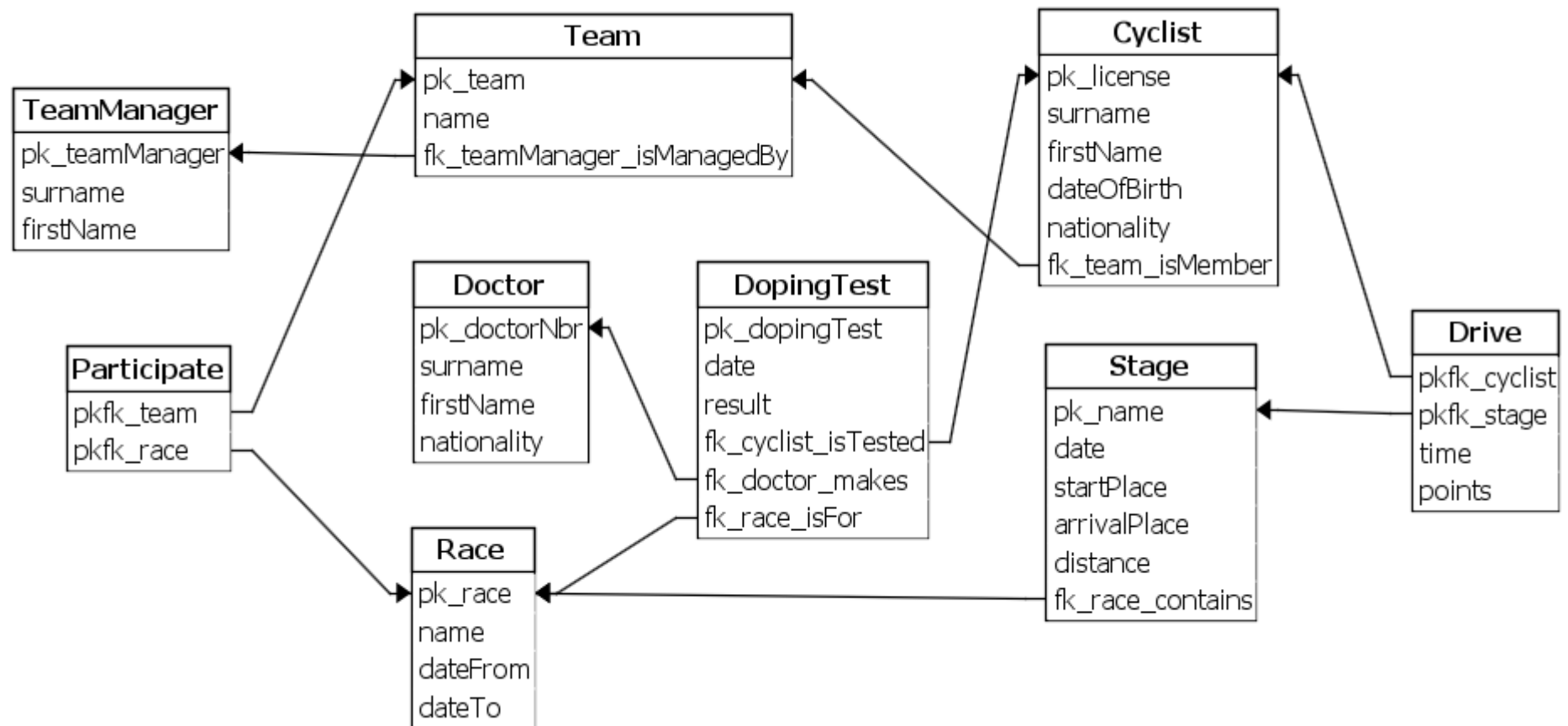


Question 1 : Elaboration MCD



Question 2 : Traduction MCD



Question 3 : Requêtes

Query A

```

1 • SELECT surname AS "Surname", firstName AS "First name", race.pk_name AS "Race name", startDate AS "Start date"
2 FROM cyclist
3 INNER JOIN participates
4     ON pk_license = pkfk_cyclist_Participates
5 INNER JOIN stage
6     ON pkfk_stage_Participates = stage.pk_code
7 INNER JOIN race
8     ON fk_race_consists = race.pk_name
9 ORDER BY surname ASC;
10 #ORDER BY firstName ASC;
11 #ORDER BY race.pk_name ASC;
12 #ORDER BY startDate ASC;

```

Result Grid  Filter Rows: Export:  Wrap Cell Content: 

	Surname	First name	Race name	Start date
▶	Armstark	Länz	AmstelGoldRace	2005-04-17
	Armstark	Länz	TourDeFrance	2005-07-02
	Besenmann	Stefan	AmstelGoldRace	2005-04-17
	Bezevedo	Jos	TourDeFrance	2005-07-02
	Eglisias	Kimi	TourDeFrance	2005-07-02
	Hoogard	Mich	AmstelGoldRace	2005-04-17
	Hoogard	Mich	TourDeFrance	2005-07-02

Result 7 ✕

Query B

```
1 • SELECT race.pk_name AS "Race name", COUNT(*) AS "Number of participants"
2 FROM race
3 INNER JOIN stage
4     ON race.pk_name = fk_race_consists
5 INNER JOIN participates
6     ON stage.pk_code = pkfk_stage_Participates
7 GROUP BY race.pk_name;
```

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content: 

	Race name	Number of participants
▶	AmstelGoldRace	4
	TourDeFrance	7

Query C

```

1 • SELECT pk_license AS "License number", fk_team_belongs AS "Team", surname AS "Surname", firstName AS "First Name",
2    nationality AS "Nationality", dateOfBirth AS "Date of birth", date AS "Date of doping test", result AS "Result"
3    FROM cyclist
4    INNER JOIN dopingtest
5      ON pk_license = fk_cyclist_undergoes;

```

Result Grid								
Filter Rows: Export:  Wrap Cell Content: 								
	License number	Team	Surname	First Name	Nationality	Date of birth	Date of doping test	Result
▶	109-60	Très-Mobile Team	Zevilla	Oscar	ESP	1976-09-29	2005-07-11	negative
	616-92	Rabobank	Gonzalez	Michel	POR	1977-12-12	2005-07-11	negative
	109-60	Très-Mobile Team	Zevilla	Oscar	ESP	1976-09-29	2005-07-23	negative
	109-60	Très-Mobile Team	Zevilla	Oscar	ESP	1976-09-29	2005-07-24	negative

Query D

```

1  SELECT pk_license AS "License number", fk_team_belongs AS "Team", surname AS "Surname", firstName AS "First Name",
2  nationality AS "Nationality", dateOfBirth AS "Date of birth", date AS "Date of doping test", result AS "Result"
3  FROM cyclist
4  INNER JOIN dopingtest
5  ON pk_license = fk_cyclist_undergoes
6  WHERE YEAR(date) = 2005;

```

Result Grid Filter Rows: Export: Wrap Cell Content:								
	License number	Team	Surname	First Name	Nationality	Date of birth	Date of doping test	Result
▶	109-60	Très-Mobile Team	Zevilla	Oscar	ESP	1976-09-29	2005-07-11	negative
	616-92	Rabobank	Gonzalez	Michel	POR	1977-12-12	2005-07-11	negative
	109-60	Très-Mobile Team	Zevilla	Oscar	ESP	1976-09-29	2005-07-23	negative
	109-60	Très-Mobile Team	Zevilla	Oscar	ESP	1976-09-29	2005-07-24	negative

Solution for several years:

```

1  SELECT pk_license AS "License number", fk_team_belongs AS "Team", surname AS "Surname", firstName AS "First Name",
2  nationality AS "Nationality", dateOfBirth AS "Date of birth", date AS "Date of doping test", result AS "Result"
3  FROM cyclist
4  INNER JOIN dopingtest
5      ON pk_license = fk_cyclist_undergoes
6  WHERE YEAR(date) = 2005
7      OR YEAR(date) = 2014
8      OR YEAR(date) = 2016
9      OR YEAR(date) = 2019;

```

Result Grid   Filter Rows: Export:  Wrap Cell Content: 								
	License number	Team	Surname	First Name	Nationality	Date of birth	Date of doping test	Result
▶	109-60	Très-Mobile Team	Zevilla	Oscar	ESP	1976-09-29	2005-07-11	negative
	616-92	Rabobank	Gonzalez	Michel	POR	1977-12-12	2005-07-11	negative
	109-60	Très-Mobile Team	Zevilla	Oscar	ESP	1976-09-29	2005-07-23	negative
	109-60	Très-Mobile Team	Zevilla	Oscar	ESP	1976-09-29	2005-07-24	negative
	276-04	Très-Mobile Team	Ülrik	Jean	ALL	1973-12-02	2014-11-11	positive
	258-61	Très-Mobile Team	Besenmann	Stefan	GER	1976-06-09	2016-11-11	positive
	276-04	Très-Mobile Team	Ülrik	Jean	ALL	1973-12-02	2019-11-11	positive

For more years: only one line needs to be added:

OR YEAR(date) = [new_year]

Query E

```

1 • SELECT pk_license AS "License number", fk_team_belongs AS "Team", surname AS "Surname", firstName AS "First Name",
2     nationality AS "Nationality", dateOfBirth AS "Date of birth", date AS "Date of doping test", result AS "Result"
3     FROM cyclist
4     LEFT JOIN dopingtest
5         ON pk_license = fk_cyclist_undergoes
6     WHERE pk_test IS NULL;

```

Result Grid   Filter Rows: Export:  Wrap Cell Content: 						
	License number	Team	Surname	First Name	Nationality	Date of birth
▶	256-89	Très-Mobile Team	Tsabel	Rikki	ALL	1970-07-07
	258-62	Disney-Channel	Yoakim	Benoa	LUX	1976-01-14
	258-66	Quick-Fall	Littini	Paolo	ITA	1977-05-12
	259-48	Rabobank	Hoogard	Pitti	NED	1978-02-24
	300-09	Rabobank	Hoogard	Mich	NED	1975-11-20
	403-24	Disney-Channel	Bezevedo	Jos	LUX	1976-08-24
	404-24	Disney-Channel	Bezevedo	Jos	POR	1973-09-19
	503-99	Team Tse-Tse	Vogel	Jens	GER	1975-05-19
	700-11	Fassa Bambino	Eglisias	Kimi	ESP	1976-03-03
	718-33	Disney-Channel	Armstark	Länz	USA	1971-09-18

Query F

```
1 • SELECT surname AS "Surname", firstName AS "First name", SUM(points) AS "Total points"
2 FROM cyclist
3 INNER JOIN participates
4     ON pk_license = pkfk_cyclist_Participates
5 GROUP BY pk_license;
```

Result Grid			
Filter Rows:			
Export:  Wrap Cell Content: 			
	Surname	First name	Total points
►	Besenmann	Stefan	74
	Littini	Paolo	68
	Hoogard	Mich	222
	Bezevedo	Jos	40
	Vogel	Jens	35
	Eglisias	Kimi	50
	Armstark	Länz	160

Query G

```

1  SELECT surname AS "Surname", firstName AS "First name", race.pk_name AS "Race name", SUM(points) AS "Total points"
2  FROM cyclist
3  INNER JOIN participates
4      ON pk_license = pkfk_cyclist_Participates
5  INNER JOIN stage
6      ON pkfk_stage_Participates = stage.pk_code
7  INNER JOIN race
8      ON fk_race_consists = race.pk_name
9  GROUP BY surname, firstName, race.pk_name
10 ORDER BY surname, firstName;

```

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content: 

	Surname	First name	Race name	Total points
▶	Armstark	Länz	AmstelGoldRace	80
	Armstark	Länz	TourDeFrance	80
	Besenmann	Stefan	AmstelGoldRace	74
	Bezevedo	Jos	TourDeFrance	40
	Eglisias	Kimi	TourDeFrance	50
	Hoogard	Mich	AmstelGoldRace	62
	Hoogard	Mich	TourDeFrance	160
	Littini	Paolo	AmstelGoldRace	68
	Vogel	Jens	TourDeFrance	35

Query H

```
1 • SELECT race.pk_name AS "Race name", surname AS "Surname", firstName AS "First name", CONCAT(SUM(points) * 3, " €") AS "Price money"
2 FROM cyclist
3 INNER JOIN participates
4     ON pk_license = pkfk_cyclist_Participates
5 INNER JOIN stage
6     ON pkfk_stage_Participates = stage.pk_code
7 INNER JOIN race
8     ON fk_race_consists = race.pk_name
9 GROUP BY surname, firstName, race.pk_name
10 ORDER BY race.pk_name, surname, firstName;
```

Result Grid   Filter Rows: | Export:  | Wrap Cell Content: 

	Race name	Surname	First name	Price money
▶	AmstelGoldRace	Armstark	Länz	240 €
	AmstelGoldRace	Besenmann	Stefan	222 €
	AmstelGoldRace	Hoogard	Mich	186 €
	AmstelGoldRace	Littini	Paolo	204 €
	TourDeFrance	Armstark	Länz	240 €
	TourDeFrance	Bezevedo	Jos	120 €
	TourDeFrance	Eglisias	Kimi	150 €
	TourDeFrance	Hoogard	Mich	480 €
	TourDeFrance	Vogel	Jens	105 €

Query I

```
1 • UPDATE team
2   SET manager = UPPER(manager);
3   #SET manager = LOWER(manager);
```

Result:

```
1 • SELECT * FROM exercise07.team;
```

Result Grid		
Filter Rows:		
Ed		
	pk_name	manager
▶	Disney-Channel	JOHANN DUCK
	Fassa Bambino	DONALD DUCK
	Quick-Fall	VON TIEF BERT
	Robobank	QUINTO JOAO
	Team Tse-Tse	VAN DER POOL LISI
	Très-Mobile Team	WALTER KNALLTER
•	NULL	NULL

Query J

```

1  UPDATE stage
2  SET distance = distance / 1.6;
3  #SET distance = distance * 1.6;

```

Result:

```
1 • SELECT * FROM exercise07.stage;
```

Result Grid						
		Filter Rows:	Edit:		Export/Import:	Wrap Cell Content:
	pk_code	fk_race_consists	date	start	arrival	distance
▶	AmstelGoldRace-01	AmstelGoldRace	2005-04-17	Maastricht	Valkenburg	112.5
	Paris-Roubaix-01	Paris-Roubaix	2005-04-10	Compiègne	Roubaix	162.5
	TourDeFrance-01	TourDeFrance	2005-07-02	Fromentine	Ile de Noirmoutier	11.875
	TourDeFrance-02	TourDeFrance	2005-07-03	Challans	Les Essarts	113.74999999999997
	TourDeFrance-03	TourDeFrance	2005-07-04	La Chataigneraie	Tours	130
•	NULL	NULL	NULL	NULL	NULL	NULL

Query K

Add new doctors

```
1 • INSERT INTO doctor(pk_code, surname, firstName)
2   VALUES("111", "aaa", "aaa"),
3   ("222", "bbb", "bbb");
```

REsult:

```
1 • SELECT * FROM exercise07.doctor;
```

Result Grid				
Filter Rows:				
Edit:   				
	pk_code	surname	firstName	nationality
▶	103	Roinard	Emilien	NULL
	105	Duclos	Pierre	NULL
	111	aaa	aaa	NULL
	222	bbb	bbb	NULL
*	NULL	NULL	NULL	NULL

Remove those doctors

```
5  #SELECT *
6  • DELETE doctor
7  FROM doctor
8  LEFT JOIN dopingtest
9      ON pk_code = fk_doctor_performs
10 WHERE fk_doctor_performs IS NULL;
```

Result:

```
1 • SELECT * FROM exercise07.doctor;
```

Result Grid

Filter Rows:

	pk_code	surname	firstName	nationality
▶	103	Roinard	Emilien	NULL
	105	Dudlos	Pierre	NULL
•	NULL	NULL	NULL	NULL